## Supply Chain Resilience & Autonomous Adaptation: Final Simulation Report

### Executive Summary

This project investigates the resilience of food supply chains under extreme climate and geopolitical disruptions. By combining discrete-event simulation (AnyLogic) with statistical modeling (Python/R) and a Multi-Agent System (LangGraph/Ollama), this study models both the physical impact of supply shocks and the potential for autonomous AI agents to drive rapid system recovery. The findings demonstrate the critical threshold between manageable bottlenecks and total system failure, highlighting the necessity of pre-programmed autonomous recovery protocols.

### Methodology

The architecture of this experiment is divided into physical simulation and autonomous logic:

1. **Physical Flow Simulation (AnyLogic):** A discrete-event model representing a three-node distribution network (Farms → Warehouse → Distribution Center → Consumers). The model incorporates production rates, transit delays, and queue capacities.
2. **Disruption Injection:** A programmable climate disruption event occurs at $t = 30$, reducing primary farm output by a variable severity parameter ($\lambda$).
3. **Autonomous Recovery Agent:** At $t = 45$, a simulated agentic response evaluates the shortage and activates a backup external supplier to restore equilibrium.
4. **Statistical Aggregation (Python/R):** Time-series inventory data is extracted and analyzed using Kruskal-Wallis tests and linear regression to quantify the relationship between disruption severity and critical stock-out durations.

### Scenario Analysis & Key Findings

#### Scenario A: Minor Disruption (20% Severity)

In a low-severity scenario ($\lambda = 0.2$), primary farm production drops slightly but still outpaces the processing capacity of the downstream transit delays.

- **System Behavior:** The network experiences a persistent bottleneck rather than a shortage. Incoming inventory exceeds the transit capacity of 150 units/day, causing stock to continuously pile up in the primary warehouse queue.
- **Impact:** Consumer fulfillment remains uninterrupted. The system demonstrates high inherent resilience, absorbing the 20% shock without ever breaching critical minimums. Autonomous recovery intervention at $t = 45$ is largely redundant in this state.

#### Scenario B: Severe Disruption (80% Severity)

In a high-severity climate shock ($\lambda = 0.8$), primary production collapses drastically, simulating a major geopolitical or environmental failure.

- **System Behavior:** The inflow rate drops below the transit capacity. Existing warehouse buffers are rapidly depleted. The system enters a flatline state at $0$ inventory, representing a total stock-out.

- **Impact:** Consumer fulfillment halts completely. The delay between the disruption ($t = 30$) and the autonomous activation of the backup supplier ($t = 45$) represents the network's maximum vulnerability window.

**Agentic AI Integration & Optimization**

The traditional simulation highlights the physical breakdown of the supply chain, but the accompanying LangChain multi-agent architecture provides the solution.

By deploying local, privacy-preserving LLMs (Llama 3.2), the system can autonomously monitor these exact stock levels in real-time. When the Monitor Agent detects the trajectory toward the $0$-stock flatline seen in Scenario B, the Replenishment Agent bypasses manual human approval delays. It mathematically calculates the exact deficit rate and instantly triggers the External Supplier to route backup shipments, actively minimizing the "Days Below Critical Stock Level."

**Conclusion**

The simulation confirms that supply chain resilience is non-linear; networks can effortlessly absorb minor shocks but fail catastrophically once a specific severity threshold is crossed. To maintain continuous, sustainable replenishment—as outlined by the EU Horizon CARES framework—modern supply chains cannot rely solely on physical buffer stock. They require the integration of Agentic AI foundations capable of autonomous adaptation, rapid rerouting, and real-time disruption mitigation.